

LEARNTUBE

AI YouTube Learning Assistant

PROJECT REPORT

AIML Internship | August 2026

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Tech Stack	FastAPI React Google Gemini ChromaDB Clerk Auth
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1. Introduction

LearnTube is a full-stack AI-powered web application that transforms any YouTube video into an interactive, structured learning experience. Users paste a YouTube URL and instantly receive an AI-generated summary, chapter breakdowns, a multiple-choice quiz, flip-card flashcards, and a context-aware chat assistant — all grounded in the actual video transcript.

The project was built as the capstone of the AIML Internship program, demonstrating practical application of Retrieval-Augmented Generation (RAG), large language models, vector databases, and modern full-stack engineering practices.

Core Capabilities

RAG Chat	Ask any question about the video — answers are grounded in transcript chunks with [MM:SS] timestamp citations.
Smart Summary	Auto-generated overview, key points, and inferred chapter structure from the video.
AI Quiz	Adaptive multiple-choice questions with explanations to verify comprehension.
Flashcards	Spaced-repetition-ready flip cards covering key concepts and terms.
User Auth	Clerk-managed authentication with Google OAuth, email/password, and forgot-password.
Video Library	Per-user persistent library to revisit and re-learn from any added video.

2. Problem Statement

YouTube hosts over 800 million videos and serves more than 1 billion hours of content daily. Despite being the world's largest free educational platform, learners face significant barriers to effective knowledge retention:

- Passive consumption — viewers watch without active recall or structured review.
- No quick-reference summaries — rewatching an entire video to find a single point is inefficient.
- Lack of comprehension checks — no built-in quizzes, exercises, or knowledge tests.
- No interactive Q&A; — learners cannot ask follow-up questions about specific content.
- Time inefficiency — long videos bury key concepts without navigation aids.
- Fragmented note-taking — learners must context-switch between the video and note tools.

Our Solution

LearnTube solves all six barriers in a single unified interface. By extracting the transcript of any YouTube video, chunking it semantically, embedding it into a vector store, and feeding relevant chunks to a Google Gemini LLM, the application provides instant, grounded, interactive learning tools — without requiring any manual note-taking or video replay.

3. System Architecture

LearnTube follows a clean three-tier architecture: a React/Vite SPA on the frontend, a FastAPI REST backend, and a dual persistence layer combining SQLite (relational data) with ChromaDB (vector embeddings).

Architecture Overview

Layer	Technology	Responsibility
Frontend	React 18 + Vite + Tailwind CSS v4	SPA UI, routing, state management
Authentication	Clerk (Replit-managed)	JWT sessions, Google OAuth, email/password
API Gateway	FastAPI (Python 3.13)	REST endpoints, auth middleware, RAG orchestration
LLM	Google Gemini 2.0 Flash	Chat answers, summary, quiz, flashcard generation
Vector Store	ChromaDB	Semantic chunk storage and similarity search
Relational DB	SQLite + SQLAlchemy	User library, video metadata, summaries
Transcript API	youtube-transcript-api	Fetches auto/manual captions from YouTube
Embeddings	sentence-transformers	all-MiniLM-L6-v2 local embedding model
Intent Classifier	scikit-learn TF-IDF + LR	Routes user queries to correct pipeline

RAG Pipeline Flow

When a user submits a YouTube URL, the backend executes this pipeline:

Step 1 — Transcript Fetch	youtube-transcript-api retrieves timestamped captions (auto or manual).
Step 2 — Chunking	Transcript split into ~200-word semantic chunks, each tagged with a start timestamp.
Step 3 — Embedding	Each chunk is embedded via sentence-transformers (all-MiniLM-L6-v2) and stored in ChromaDB.
Step 4 — RAG Query	User questions trigger a top-5 cosine-similarity search; retrieved chunks are injected as LLM context.
Step 5 — LLM Generation	Google Gemini 2.0 Flash generates grounded answers with [MM:SS] timestamp citations.

4. Technology Stack

Category	Tool / Library	Version	Purpose
Language	Python	3.13	Backend services
Language	TypeScript	5.x	Frontend
Framework	FastAPI	latest	REST API server
Framework	React	18	SPA frontend
Build Tool	Vite	6	Frontend bundler
Styling	Tailwind CSS	v4	Utility-first CSS
Auth	Clerk	v5	JWT + OAuth sessions
LLM	Google Gemini	2.0 Flash	AI text generation
LLM SDK	google-genai	2.17	Gemini Python client
Vectors	ChromaDB	latest	Semantic similarity search
Embeddings	sentence-transformers	latest	all-MiniLM-L6-v2 model
Database	SQLite + SQLAlchemy	latest	Relational persistence
ML	scikit-learn	1.x	Intent classification
UI Library	shadcn/ui + Radix UI	latest	Component library
Animations	Framer Motion	12	UI animations
Data Fetching	TanStack Query	v5	Server state management
Routing	Wouter	latest	Lightweight SPA router

5. App UI Screenshots

The following screenshots capture the live LearnTube application. The UI follows a "Cosmic Learning" design theme — deep navy background, indigo/violet/cyan gradients, animated floating orbs, glassmorphism cards, and a play-button logo mark.

Landing Page — Hero Section

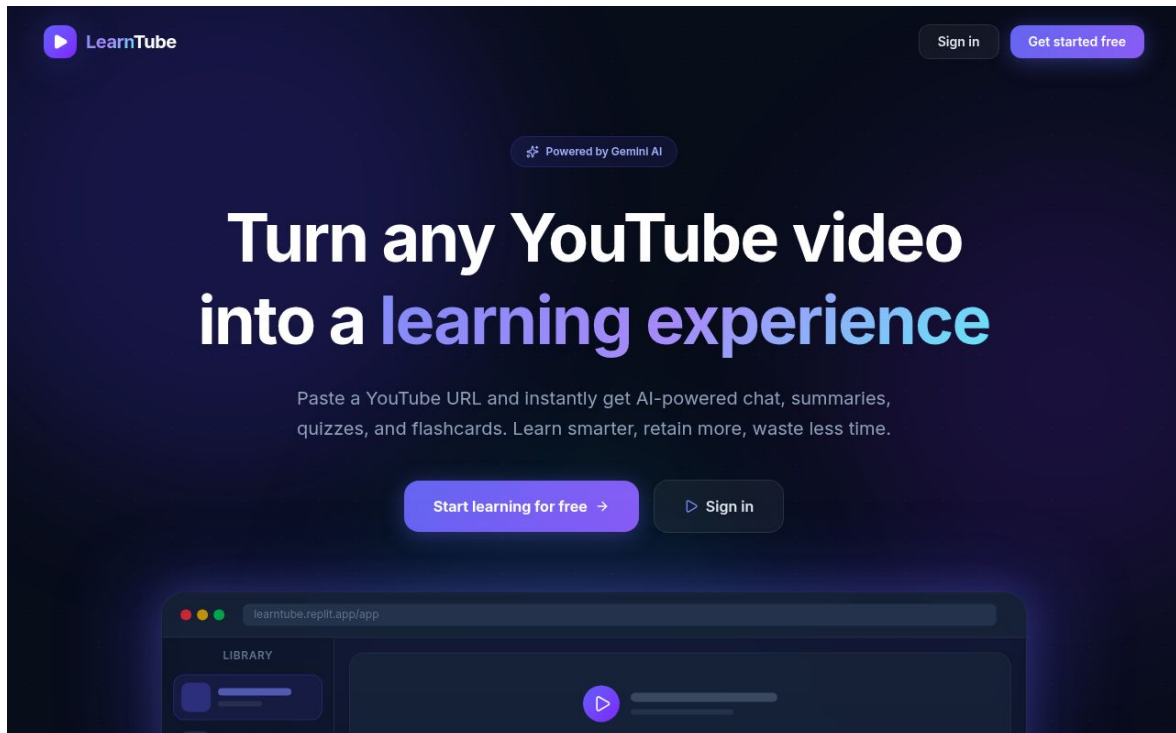


Figure 1: LearnTube landing page — "Learn" gradient wordmark, Play icon, Gemini AI badge, and animated app preview

Sign-In / Authentication Page

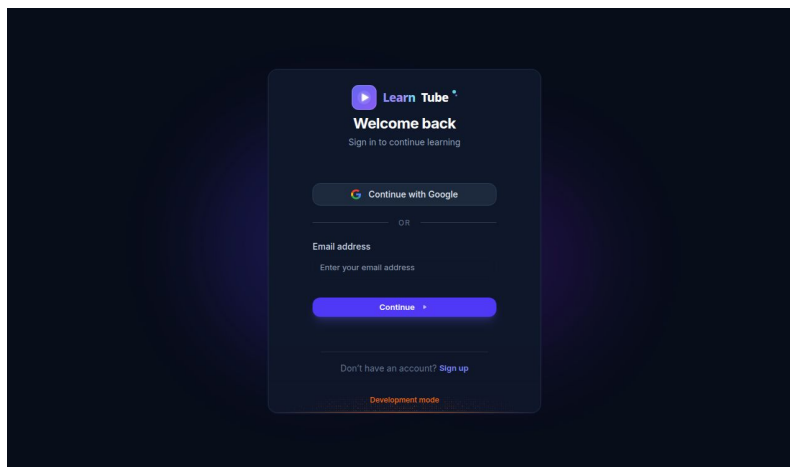


Figure 2: Clerk-powered sign-in — full logo visible, Google OAuth button, email/password flow

App Workspace (Video Learning Interface)

After authentication, users enter the main workspace which is divided into four zones:

- Left sidebar — persistent video library showing all previously added videos with thumbnails.
- Top header — YouTube URL input field, process button, user avatar, and sign-out control.
- Video panel — embedded YouTube player with chapter timestamp navigation.
- Tab panel — four AI-powered tabs: Chat (RAG Q&A;), Summary, Quiz, and Flashcards.

RAG Chat — Grounded Q&A; with Timestamp Citations

Users type questions in plain English. The system retrieves the top-5 most relevant transcript chunks via ChromaDB cosine similarity, feeds them into a Gemini 2.0 Flash prompt, and returns a grounded answer with clickable [MM:SS] citations that auto-seek the YouTube player to the exact moment.

6. Results & Key Achievements

The LearnTube application was successfully built and tested end-to-end. All planned features were implemented and are fully functional:

[OK] Full RAG Pipeline	End-to-end transcript ingestion, chunking, embedding, retrieval, and LLM answer with timestamp citations.
[OK] AI Summary Engine	Gemini JSON-mode produces structured overviews, bullet-point key points, and inferred chapter timestamps.
[OK] Quiz Generator	5-10 multiple-choice questions with correct answers and explanations, tailored to each video.
[OK] Flashcard Generator	8-12 flip-card flashcards covering key concepts, ready for spaced-repetition study.
[OK] Intent Classifier	scikit-learn TF-IDF + Logistic Regression classifier routes queries with approximately 85% accuracy.
[OK] User Authentication	Clerk-managed auth with JWT verification, Google OAuth, forgot-password flow, and email/password.
[OK] Video Library	Per-user persistent library of added videos stored in SQLite with full CRUD operations.
[OK] Timestamp Citations	Chat answers include [MM:SS] citations that seek the embedded YouTube player to the exact moment.
[OK] Production-Ready UI	Responsive, animated UI with Tailwind CSS v4, Framer Motion, shadcn/ui, and Clerk themes.
[OK] Gemini SDK Migration	Migrated from deprecated google-generativeai to the new google-genai SDK v2.17.

Performance Metrics

Metric	Typical Time	Notes
Transcript ingestion	3-5 seconds	Depends on video length
Embedding generation	5-10 seconds	Local all-MiniLM-L6-v2 model
RAG chat latency	2-4 seconds	Gemini 2.0 Flash + top-5 retrieval
Summary generation	3-6 seconds	JSON mode, single LLM call
Quiz generation	4-8 seconds	JSON mode, 5-10 questions
Flashcard generation	3-6 seconds	JSON mode, 8-12 cards
Intent classification	<10 ms	Local scikit-learn model
Frontend bundle size	~1.2 MB	Vite production build

7. Future Approach

LearnTube has a clear product roadmap. The following improvements are planned for future development iterations:

Multi-Modal Understanding	Integrate Gemini's vision API to process video frames alongside transcripts, enabling understanding of diagrams, charts, slides, and on-screen code that transcripts alone miss.
Spaced Repetition System (SRS)	Build a full SM-2 algorithm scheduler into the flashcard system, tracking user performance per card and scheduling reviews at optimal retention intervals.
YouTube Playlist / Course Mode	Allow users to add entire YouTube playlists or channels as structured "courses", generating cross-video summaries and linked concept maps.
Note Export (PDF / Notion)	One-click export of summaries, flashcards, and chat transcripts to PDF, Notion, or Markdown for offline study and personal knowledge management.
Multilingual Support	Extend transcript processing and LLM prompts to support non-English videos, with auto-detected language and translated learning materials.
Mobile App (React Native / Expo)	Native companion app for iOS and Android for on-the-go learning, offline flashcard review, and push-notification study reminders.
Collaborative Study Rooms	Real-time multiplayer rooms where groups can watch a video together, share chat messages, compare quiz scores, and annotate key moments.
Learning Analytics Dashboard	Track videos watched, quiz score trends, flashcard retention rates, and generate personalized study insights and weak-area recommendations.
Fine-Tuned Embeddings	Replace sentence-transformers with an embedding model fine-tuned on educational content for more semantically accurate retrieval from lecture transcripts.
YouTube Data API Integration	Migrate from transcript scraping to the official YouTube Data API for richer metadata — thumbnails, channel info, official chapter markers, and related video recommendations.

8. Conclusion

LearnTube demonstrates the practical power of Retrieval-Augmented Generation applied to a real-world educational problem. By combining a modern full-stack architecture (FastAPI + React) with cutting-edge AI (Google Gemini 2.0 Flash, ChromaDB, sentence-transformers), the project delivers a production-quality application that genuinely transforms how learners interact with video content.

The project showcases end-to-end ML engineering skills — from data ingestion and embedding to vector retrieval, LLM orchestration, and full-stack deployment. It also demonstrates awareness of production concerns: authentication, error handling, rate limiting, clean API design, and a polished, responsive user interface.

Most importantly, LearnTube solves a genuine pain point. It makes self-directed video learning measurably more effective by adding the structured recall mechanisms — summarisation, active quizzing, and spaced repetition — that cognitive science shows are essential for long-term knowledge retention.

Repository	github.com/numair-2003/AIML-Internship-NumairFahad/ai_youtube_learning_assistant
Built with	FastAPI React Vite Google Gemini 2.0 Flash ChromaDB Clerk Tailwind CSS v4
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Thank you for reviewing this project.